

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. **(BL4)**

Assessment Tool Weightage: 50%

Time Duration: 30 Minutes

Questions: 30

Total Marks: 30

Annexure A

MCQ

Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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CSA17 — ARTIFICIAL INTELLIGENCE

Assessment Tool 3 — MCQ: Answer Key with Explanations

CO3: Knowledge representation and reasoning — propositional logic, FOL, inference, resolution, chaining, learning

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
- b) An agent that uses a knowledge base and logical inference to decide actions ✓**
- c) An agent that only follows pre-defined scripts
- d) An agent that learns from rewards

Why: A logical agent maintains a knowledge base of facts/rules and derives new conclusions through logical inference to decide its actions, rather than reacting randomly, following fixed scripts, or learning purely from reward signals.

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
- b) $\neg P \vee Q$
- c) $\neg Q \rightarrow \neg P$ only
- d) Both b and c ✓**

Why: $P \rightarrow Q$ is equivalent both to its material-implication form $\neg P \vee Q$ and to its contrapositive $\neg Q \rightarrow \neg P$ — both hold simultaneously, so 'both b and c' is the complete correct answer.

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
- b) A conjunction of disjunctions (clauses) ✓**
- c) A conjunction of literals only
- d) A sequence of implications

Why: CNF is, by definition, an AND of ORs — a conjunction of clauses, where each clause is itself a disjunction of literals.

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
- b) Every entity that is a student passes ✓**
- c) If something passes, it is a student
- d) There exists at least one student who passes

Why: The universal quantifier applied to an implication states that for every x , being a student implies passing — i.e., every student passes (it says nothing about non-students).

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical ✓**
- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

Why: Unification searches for a substitution of variables that makes two atomic sentences match exactly, which is a prerequisite step used inside resolution and chaining, not those procedures themselves.

Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

- a) The knowledge base is consistent
- b) The negated goal is satisfiable
- c) A contradiction — the original goal is proven ✓**

d) More clauses need to be added

Why: The empty clause represents a logical contradiction (False), which arises exactly when $KB \wedge \neg \text{Goal}$ is unsatisfiable — proving the original goal follows from the KB (resolution refutation).

Q7. Forward Chaining is best described as a _____ strategy.

a) Goal-driven / top-down

b) Data-driven / bottom-up ✓

c) Heuristic-based depth-first

d) Probabilistic inference

Why: Forward chaining starts from known facts and repeatedly applies rules to derive new facts, working outward from the data until the goal is (possibly) reached — a bottom-up, data-driven process.

Q8. Backward Chaining begins reasoning from:

a) All known facts toward a conclusion

b) The goal and works backward to verify supporting facts ✓

c) A random clause in the knowledge base

d) The resolution refutation tree

Why: Backward chaining starts from the goal to be proven and works backward, checking whether the facts/rules needed to support it are true — the opposite direction from forward chaining.

Q9. Which step is NOT part of converting a sentence to CNF?

a) Eliminate biconditionals ($\leftrightarrow$)

b) Move negations inward using De Morgan's laws

c) Apply Skolemization to remove existential quantifiers

d) Distribute AND over OR ✓

Why: CNF requires distributing OR over AND ($A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$) to reach a conjunction of disjunctions. Distributing AND over OR instead produces DNF, not CNF, so it is not a genuine CNF-conversion step.

Q10. Knowledge Engineering in FOL involves:

a) Training a neural network on domain-specific data

b) Defining domain concepts, relations, and axioms formally in FOL ✓

c) Writing heuristic search strategies

d) Encoding reward functions for reinforcement learning

Why: Knowledge engineering is the process of formally identifying and encoding a domain's objects, relations, and rules as FOL sentences so a reasoning system can use them.

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

a) $\neg Q$ is true

b) Q must be true ✓

c) P is uncertain

d) The KB is inconsistent

Why: Modus Ponens is the classic inference rule: from $P \rightarrow Q$ and P, Q can be validly concluded.

Q12. The Most General Unifier (MGU) is the substitution that:

a) Is the most specific possible match

b) Makes two terms identical with minimal variable binding ✓

c) Eliminates all variables from a clause

d) Applies resolution to produce CNF

Why: The MGU is the least-committal substitution that still makes two expressions match — it binds only what is necessary, keeping the unification as general as possible.

Q13. Which inference method is more appropriate when a specific goal is known?

- a) Forward Chaining
- b) Truth Table method
- c) Backward Chaining ✓**
- d) A* search

Why: Backward chaining works directly from a known goal back to supporting facts, making it far more efficient than data-driven forward chaining when the goal is already specified.

Q14. Propositional Logic differs from First Order Logic because it:

- a) Supports probabilistic reasoning
- b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions ✓**
- c) Is always more expressive
- d) Handles uncertainty better

Why: Propositional logic only assigns truth values to whole atomic statements; it has no way to represent individual objects, their relations, or quantify over them the way FOL does.

Q15. Resolution Refutation proves a goal G by:

- a) Directly deriving G from the KB using forward chaining
- b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$) ✓**
- c) Checking G in the truth table
- d) Applying Modus Tollens repeatedly

Why: Resolution refutation is a proof-by-contradiction technique: negate the goal, add it to the KB, and show the result is unsatisfiable (derives the empty clause), which proves the original goal is entailed.

Q16. Learning from observation in AI refers to:

- a) An agent receiving numerical rewards from the environment
- b) Improving performance by examining examples or experience from the environment ✓**
- c) Acquiring knowledge by following explicit symbolic rules
- d) Memorizing a fixed dataset without generalization

Why: Learning from observation broadly means an agent improves its behaviour/knowledge by examining examples or experience, rather than being explicitly programmed with rules or only memorizing data.

Q17. Inductive learning is best described as:

- a) Deriving specific conclusions from general rules
- b) Generalizing a hypothesis from specific training examples ✓**
- c) Memorizing training data exactly
- d) Applying search algorithms to find optimal policies

Why: Induction moves from specific observed examples to a general hypothesis that (hopefully) also holds for unseen data — the opposite direction of deduction.

Q18. In a Decision Tree, what does each internal node represent?

- a) A class label / output decision
- b) A probability distribution over outcomes
- c) A test on an attribute / feature ✓**
- d) A training data sample

Why: Internal nodes in a decision tree represent a test/split on a specific attribute; the leaves (not internal nodes) hold the final class label/decision.

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

- a) Requires large amounts of training data to generalize

- b) Learns only from numeric data using regression
- c) Uses prior domain knowledge to explain and generalize a single example ✓**
- d) Applies unsupervised clustering to find hidden patterns

Why: EBL uses existing domain theory to construct an explanation of why one example is an instance of a concept, then generalizes that explanation — needing far fewer examples than pure inductive learning.

Q20. In statistical learning, a Naive Bayes classifier assumes:

- a) All features are conditionally dependent on each other given the class
- b) The data must follow a uniform distribution
- c) All features are conditionally independent given the class label ✓**
- d) Labels are determined by a reward function

Why: The 'naive' assumption is that all features are conditionally independent of each other given the class label, which simplifies the probability computation considerably.

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

- a) Using the EM algorithm to estimate missing values
- b) Training a deep neural network with dropout regularization
- c) Directly computing maximum likelihood estimates from observed data ✓**
- d) Applying reinforcement signals to update weights

Why: When every variable is observed (no missing/hidden data), the network's parameters can be estimated directly via maximum likelihood from the observed counts/frequencies — no need for EM.

Q22. The Expectation-Maximization (EM) algorithm is used when:

- a) Training data is fully labeled and no variables are hidden
- b) An agent explores an unknown environment using trial and error
- c) The dataset contains hidden (latent) variables that cannot be directly observed ✓**
- d) Backpropagation fails to converge in a neural network

Why: EM is specifically designed for parameter estimation when the data has hidden/latent variables, iterating between estimating those hidden values (E-step) and re-fitting parameters (M-step).

Q23. In a Neural Network, the activation function serves to:

- a) Initialize the weights of each neuron to zero
- b) Normalize the input dataset before training
- c) Introduce non-linearity to enable learning of complex patterns ✓**
- d) Store previously learned examples in memory

Why: Without a non-linear activation function, stacking layers would collapse into an equivalent single linear transformation; activation functions are what let networks model complex, non-linear relationships.

Q24. Which of the following correctly describes backpropagation in a neural network?

- a) Weights are updated in the forward pass using input gradients
- b) Outputs of each layer are stored and reused in later epochs
- c) The error is propagated backward from output to input layers to adjust weights ✓**
- d) A separate model is trained for each hidden layer independently

Why: Backpropagation computes the output error and propagates it backward through the network layer by layer, using the chain rule to compute gradients that update each layer's weights.

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

- a) Minimizing the number of training examples used
- b) Grouping similar data points into clusters using centroids
- c) Finding the maximum-margin hyperplane that separates classes ✓**

d) Updating weights using Q-values from an environment

Why: SVMs find the hyperplane that separates classes while maximizing the margin (distance) to the nearest data points from each class (the support vectors).

Q26. The kernel trick in kernel machines (SVMs) allows:

a) Reducing the number of support vectors needed

b) Accelerating gradient descent convergence

c) Operating in a high-dimensional feature space without explicitly computing coordinates ✓

d) Replacing hidden layers in a deep neural network

Why: The kernel trick computes inner products in a high-dimensional (even infinite-dimensional) feature space using a kernel function, avoiding the cost of ever explicitly mapping data into that space.

Q27. In Reinforcement Learning, the agent aims to maximize:

a) Classification accuracy on a fixed test set

b) The number of features selected from the training data

c) Cumulative reward received over time through interaction with the environment ✓

d) The size of the neural network hidden layer

Why: RL's objective is defined entirely around maximizing the total (often discounted) reward the agent accumulates through its interactions with the environment, not a fixed labeled-accuracy metric.

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

a) A supervised learning algorithm requiring labeled datasets

b) An unsupervised clustering technique using centroids

c) A model-free, off-policy temporal difference learning method ✓

d) A decision-tree-based ensemble classifier

Why: Q-Learning learns action-values directly from experience without needing a model of the environment (model-free), and it learns about the optimal policy while following a possibly different exploratory policy (off-policy), using temporal-difference updates.

Q29. Overfitting in machine learning occurs when:

a) The model performs poorly on both training and test data

b) The learning rate is set too low during gradient descent

c) The model fits training data too closely and generalizes poorly to new data ✓

d) The training set has more features than training examples

Why: Overfitting is specifically when a model captures noise/idiosyncrasies of the training set so closely that it performs well there but fails to generalize to new, unseen data.

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

a) RL uses labeled input-output pairs; SL uses reward signals

b) Both RL and SL require explicit reward functions

c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples ✓

d) RL is always faster to converge than SL on any dataset

Why: The fundamental distinction is the learning signal: RL learns through trial-and-error interaction guided by reward feedback, while SL learns from a fixed set of labeled input-output examples.

Quick-Reference Answer Key

Q#	Ans	Q#	Ans	Q#	Ans
1	B	11	B	21	C
2	D	12	B	22	C
3	B	13	C	23	C
4	B	14	B	24	C
5	B	15	B	25	C
6	C	16	B	26	C
7	B	17	B	27	C
8	B	18	C	28	C
9	D	19	C	29	C
10	B	20	C	30	C